

Meeting Summaries of December 4

Date: December 4th

Time: 1:00–1:25 PM(PDT)

Attendees: Prof. Jan, Dr. Javed, Dr. Yue and Feng

Topic: Feng majorly reported his progress on clustering analysis

Summary of Discussion

- Feng started the clustering analysis with pollution scores from the PCA assessment, mainly followed the steps in the 3 and 4 chapters of the book "Numerical Ecology with R" by Legendre and Legendre (2019).
- Feng reported the three core steps of clustering analysis:
 1. Set up proper dissimilarity(distance) measure for species abundance data (*composition* vs *abundance*)
 2. (Internal validation) Compare the performance of different linkage functions in terms of dissimilarity retained between pairs across resulted clusters. It validates whether the meta dissimilarity information is kept well in the clustering process.
 3. (External validation) Compare the clustering results with externally independent information, e.g., habitat features. Check how the species clusters are ecologically interpretable with respect to these interested information.
- Feng briefly outlined the simulation method in testing hypotheses power on a mean regression model. These shared ideas will be transferred to the quantile regression model to support hypotheses power tests soon.
- Prof. Jan suggested Feng to duplicate Jian's steps in clustering analysis, using the same dissimilarity measures (octave scale + Euclidean distance) and linkage function (Ward's method) as a benchmark. Additionally, include the water velocity into the ANOVA with habitat features where the velocity in lakes be assumed as zero.
- Prof. Javed recommended Feng to summarize his interaction notes with three mentors, like the tasks completed, course(s) taken and other details, so that they can write a good recommendation letter for Feng's PhD applications.
- Prof. Javed suggested Feng to use the full data set as soon as possible, so that the analysis results will be more stable and reliable.

Action Items

- Feng will first duplicate Jian's clustering analysis steps and add the water velocity variable into the ANOVA with habitat features.
- Feng will summarize his interactions with three mentors to make reference letters easier to write, three individual summaries (within one page) to each mentor.
- Feng will build the Linear Discriminant Analysis (LDA) model based on the clustering results and report it in the next meeting.

Follow-Up

Temporary meeting time: December 11th, 1:00–1:25 PM(PDT)

Supportive materials

- A meeting recording with subtitles has been restored at Feng's laptop.